

PROBLEM SET 1

Transport Phenomena I
ChE 311

Transport Phenomena 2nd Edition

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Using the unit vectors of a Cartesian coordinate system, prove the following useful identities:

- a- $\text{div}(\rho \langle v |) = \langle \nabla \rho | v \rangle + \rho \text{div} \langle v |$
- b- $\text{div} |v \rangle \langle w | = | \nabla v | w \rangle + \langle \text{div } w | v \rangle$
- c- $\text{div} \langle v | \tau | = | \tau | : | \nabla v | + \langle v | \text{div } \tau \rangle$

Hint: use the product rule for differentiation: $\frac{\partial(fg)}{\partial x} = g \frac{\partial f}{\partial x} + f \frac{\partial g}{\partial x}$

EXERCISES

1. Write out the following summations:

(a) $\sum_{k=1}^3 k^2$ (b) $\sum_{k=1}^3 a_k^2$ (c) $\sum_{j=1}^3 \sum_{k=1}^3 a_{jk} b_{kj}$ (d) $\left(\sum_{j=1}^3 a_j\right)^2 = \sum_{j=1}^3 \sum_{k=1}^3 a_j a_k$

2. A vector $\mathbf{v}$ has components $v_x = 1$, $v_y = 2$, $v_z = -5$. A vector $\mathbf{w}$ has components $w_x = 3$, $w_y = -1$, $w_z = 1$. Evaluate:

- (a) $(\mathbf{v} \cdot \mathbf{w})$
 (b) $[\mathbf{v} \times \mathbf{w}]$
 (c) The length of $\mathbf{v}$
 (d) $(\delta_1 \cdot \mathbf{v})$
 (e) $[\delta_1 \times \mathbf{w}]$
 (f) ϕ_{vw}
 (g) $[\mathbf{r} \times \mathbf{v}]$, where $\mathbf{r}$ is the position vector.

$$1a - \sum_{k=1}^3 |k|^2 = (1)^2 + (2)^2 + (3)^2 = 1 + 4 + 9 = 14$$

$$1b - \sum_{k=1}^3 a_k^2 = (a_1)^2 + (a_2)^2 + (a_3)^2$$

$$1c - \sum_{j=1}^3 \left[\sum_{k=1}^3 a_{jk} b_{kj} \right] = (a_{1,1} \cdot b_{1,1}) + (a_{1,2} \cdot b_{2,1}) + (a_{1,3} \cdot b_{3,1}) \\ + [(a_{2,1} \cdot b_{1,2}) + (a_{2,2} \cdot b_{2,2}) + (a_{2,3} \cdot b_{3,2})] \\ + [(a_{3,1} \cdot b_{1,3}) + (a_{3,2} \cdot b_{2,3}) + (a_{3,3} \cdot b_{3,3})]$$

$$1d - \left(\sum_{j=1}^3 a_j \right)^2 = \sum_{j=1}^3 \left[\sum_{k=1}^3 a_j a_k \right] \\ = (a_1 \cdot a_1) + (a_1 \cdot a_2) + (a_1 \cdot a_3) \\ + [(a_2 \cdot a_1) + (a_2 \cdot a_2) + (a_2 \cdot a_3)] \\ + [(a_3 \cdot a_1) + (a_3 \cdot a_2) + (a_3 \cdot a_3)]$$

$$2. \quad \mathbf{v} = \langle 1, 2, -5 \rangle$$

$$\mathbf{w} = \langle 3, -1, 1 \rangle$$

$$2a - (\mathbf{v} \cdot \mathbf{w}) = (1 \cdot 3) + (-1 \cdot 2) + (-5 \cdot 1) \\ = 3 - 2 - 5 = -4$$

$$2b - [\mathbf{v} \times \mathbf{w}] = \begin{vmatrix} i & j & k \\ 1 & 2 & -5 \\ 3 & -1 & 1 \end{vmatrix} = \langle (2 \cdot 1 - (-5 \cdot -1))i, -((1 \cdot 1) - (-5 \cdot 3))j, (1 \cdot -1) - (2 \cdot 3)k \rangle \\ = \langle -3i, -16j, -7k \rangle \\ = \langle -3, -16, -7 \rangle$$

$$2c - \text{length of } \mathbf{v} = \sqrt{(1)^2 + (2)^2 + (-5)^2} = \sqrt{1 + 4 + 25} = \sqrt{30}$$

$$2d - \delta_1 \cdot \mathbf{v} = (1 \cdot 1) + (0 \cdot 2) + (0 \cdot -5) = 1$$

$$2e - [\delta_1 \times \mathbf{w}] = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 3 & -1 & 1 \end{vmatrix} = \langle 0, -(1 \cdot 0), -1 \cdot 0 \rangle \\ = \langle 0, -1, -1 \rangle$$

$$2f - \phi_{vw} = \\ (\mathbf{v} \cdot \mathbf{w}) = -4$$

$$|\mathbf{v}| = \sqrt{1^2 + 2^2 + 5^2} = \sqrt{1 + 4 + 25} = \sqrt{30}$$

$$|\mathbf{w}| = \sqrt{3^2 + (-1)^2 + (-1)^2} = \sqrt{11}$$

$$\cos \phi = \frac{-4}{\sqrt{30} \cdot \sqrt{11}} \rightarrow \phi_{vw} = \cos^{-1} \left(\frac{-4}{\sqrt{30} \cdot \sqrt{11}} \right) = 102.72^\circ$$

$$2g - [\mathbf{r} \times \mathbf{v}] = \begin{vmatrix} i & j & k \\ x & y & z \\ 1 & 2 & -5 \end{vmatrix} = \langle -5y - 2z, (-5x - z), (2x - y) \rangle \\ = \langle -5y - 2z, 5x + z, 2x - y \rangle$$

1. The components of a symmetric tensor τ are

$$\begin{array}{lll} \tau_{xx} = 3 & \tau_{xy} = 2 & \tau_{xz} = -1 \\ \tau_{yx} = 2 & \tau_{yy} = 2 & \tau_{yz} = 1 \\ \tau_{zx} = -1 & \tau_{zy} = 1 & \tau_{zz} = 4 \end{array}$$

The components of a vector $\mathbf{v}$ are

$$v_x = 5 \quad v_y = 3 \quad v_z = -2$$

Evaluate

$$\begin{array}{lll} \text{(a)} [\tau \cdot \mathbf{v}] & \text{(b)} [\mathbf{v} \cdot \tau] & \text{(c)} (\tau : \tau) \\ \text{(d)} (\mathbf{v} \cdot [\tau \cdot \mathbf{v}]) & \text{(e)} \mathbf{v} \mathbf{v} & \text{(f)} [\tau \cdot \delta_i] \end{array}$$

2. Evaluate

$$\begin{array}{ll} \text{(a)} [[\delta_1 \delta_2 \cdot \delta_2] \times \delta_1] & \text{(c)} (\delta : \delta) \\ \text{(b)} (\delta : \delta_1 \delta_2) & \text{(d)} \{\delta \cdot \delta\} \end{array}$$

$$\tau = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \quad \mathbf{v} = \langle 5, 3, -2 \rangle$$

$$\begin{aligned} \text{1a} - [\tau \cdot \mathbf{v}] &= \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \langle (3 \cdot 5) + (2 \cdot 3) + (-1 \cdot -2), (2 \cdot 5) + (2 \cdot 3) + (-1 \cdot -2), (-1 \cdot 5) + (1 \cdot 3) + (4 \cdot -2) \rangle \\ &= \langle 15 + 6 + 2 = 23, 10 + 6 + 2 = 14, -5 + 3 - 8 = -10 \rangle \\ \mathbf{v}^T &= \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 23 \\ 14 \\ -10 \end{bmatrix} = \langle 23, 14, -10 \rangle \end{aligned}$$

$$\begin{aligned} \text{1b} - [5 \ 3 \ -2] \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} &= \langle 5 \cdot 3 + 3 \cdot 2 + (-2) \cdot (-1), 5 \cdot 2 + 3 \cdot 2 + (-2) \cdot 1, -5 + 3 - 8 \rangle \\ &= \langle 15 + 6 + 2 = 23, 10 + 6 - 2 = 14, -5 + 3 - 8 = -10 \rangle \\ &= \langle 23, 14, -10 \rangle \end{aligned}$$

$$\begin{aligned} \text{1c} \quad \tau : \tau &= \sum_{i=1}^3 \sum_{j=1}^3 (\tau_{ij})^2 = (\tau_{1,1})^2 + (\tau_{1,2})^2 + (\tau_{1,3})^2 + (\tau_{2,1})^2 + (\tau_{2,2})^2 + (\tau_{2,3})^2 + (\tau_{3,1})^2 + (\tau_{3,2})^2 + (\tau_{3,3})^2 \\ &= (3)^2 + (2)^2 + (-1)^2 + (2)^2 + (2)^2 + (1)^2 + (-1)^2 + (1)^2 + (4)^2 = 9 + 4 + 1 + 4 + 4 + 1 + 1 + 1 + 16 = 41 \\ * \tau &= \begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} \end{aligned}$$

$$x, y, z = 1, 2, 3$$

$$\text{1f} - [\tau \cdot \delta_i] = \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \text{1d} - [\mathbf{v} \cdot (\tau \cdot \mathbf{v})] &= [\mathbf{v} \cdot \langle 23, 14, -10 \rangle] \\ &= \langle 5, 3, -2 \rangle \cdot \langle 23, 14, -10 \rangle \\ &= 23 \cdot 5 + 14 \cdot 3 + (-2) \cdot (-10) = 177 \\ &= \langle 3, 2, 1 \rangle \end{aligned}$$

$$\text{1e} - \mathbf{v} \mathbf{v} = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} [5 \ 3 \ -2] = \begin{bmatrix} 25 & 15 & -10 \\ 15 & 9 & -6 \\ -10 & -6 & 4 \end{bmatrix}$$

$$2a - [[\delta_i \delta_j \cdot \delta_i] \times \delta_j]$$

$$\begin{aligned} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} [0 \ 1 \ 0] &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \langle 0 + 1 + 0, 0 + 0 + 0, 0 + 0 + 0 \rangle \\ &= \langle 1, 0, 0 \rangle \times \langle 1, 0, 0 \rangle \\ &= \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{vmatrix} = \langle 0, 0, 0 \rangle \end{aligned}$$

$$2b - (\delta : \delta \delta_i) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{matrix} 0+0+0 \\ +0+0+0 \\ +0+0+0 \end{matrix} = 0$$

$$2c - (\delta : \delta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} : \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{matrix} 1^2 + 0 + 0 \\ + 0 + 1^2 + 0 \\ + 0 + 0 + 1^2 \end{matrix} = 3$$

$$2d - \{\delta \cdot \delta\} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. A field $\mathbf{v}(x, y, z)$ is said to be *irrotational* if $[\nabla \times \mathbf{v}] = 0$. Which of the following fields are irrotational?

- (a) $v_x = by$ $v_y = 0$ $v_z = 0$
 (b) $v_x = bx$ $v_y = 0$ $v_z = 0$
 (c) $v_x = by$ $v_y = bx$ $v_z = 0$
 (d) $v_x = -by$ $v_y = bx$ $v_z = 0$

3. Evaluate $(\nabla \cdot \mathbf{v})$, $\nabla \mathbf{v}$, and $[\nabla \cdot \mathbf{v}\mathbf{v}]$ for the four fields in Exercise 2.

3a. $\vec{v}_1 = \langle by, 0, 0 \rangle$ $\vec{v}_2 = \langle bx, 0, 0 \rangle$
 $\vec{v}_3 = \langle by, bx, 0 \rangle$ $\vec{v}_4 = \langle -by, bx, 0 \rangle$

$$\nabla \cdot \mathbf{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$3a(2a) - \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$3a(2b) - \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = b$$

$$3a(2c) - \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$3a(2d) - \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} = 0$$

$$3b(2a) \vec{v}_1 = \langle by, 0, 0 \rangle$$

$$\mathbf{v} = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$3b(2b) - \vec{v}_2 = \langle bx, 0, 0 \rangle$$

$$\nabla \mathbf{v} = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} b & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$3b(2c) - \vec{v}_3 = \langle by, bx, 0 \rangle$$

$$\nabla \mathbf{v} = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$3b(2d) - \vec{v}_4 = \langle -by, bx, 0 \rangle$$

$$\nabla \mathbf{v} = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$3c(2a) - \langle by, 0, 0 \rangle$$

$$\mathbf{v}\mathbf{v} = \begin{bmatrix} by & 0 & 0 \end{bmatrix} \begin{bmatrix} by \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} b^2 y^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot \mathbf{v}\mathbf{v} = \begin{bmatrix} \frac{\partial (by \cdot by)}{\partial x} + \frac{\partial (by \cdot 0)}{\partial y} + \frac{\partial (by \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot by)}{\partial x} + \frac{\partial (0 \cdot 0)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot by)}{\partial x} + \frac{\partial (0 \cdot 0)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \end{bmatrix} = \begin{bmatrix} 0+0+0 \\ 0+0+0 \\ 0+0+0 \end{bmatrix} = \langle 0, 0, 0 \rangle$$

$$3c(2b) \vec{v}_2 = \langle bx, 0, 0 \rangle$$

$$\mathbf{v}\mathbf{v} = \begin{bmatrix} bx & 0 & 0 \end{bmatrix} \begin{bmatrix} bx \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} b^2 x^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot \mathbf{v}\mathbf{v} = \begin{bmatrix} \frac{\partial (bx \cdot bx)}{\partial x} + \frac{\partial (bx \cdot 0)}{\partial y} + \frac{\partial (bx \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot bx)}{\partial x} + \frac{\partial (0 \cdot 0)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot bx)}{\partial x} + \frac{\partial (0 \cdot 0)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \end{bmatrix} = \begin{bmatrix} 2b^2 x + 0 + 0 \\ 0 + 0 + 0 \\ 0 + 0 + 0 \end{bmatrix} = \langle 2b^2 x, 0, 0 \rangle$$

$$3c(2c) \vec{v}_3 = \langle by, bx, 0 \rangle$$

$$\mathbf{v}\mathbf{v} = \begin{bmatrix} by & bx & 0 \end{bmatrix} \begin{bmatrix} by \\ bx \\ 0 \end{bmatrix} = \begin{bmatrix} b^2 y^2 & b^2 xy & 0 \\ b^2 xy & b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot \mathbf{v}\mathbf{v} = \begin{bmatrix} \frac{\partial (by \cdot by)}{\partial x} + \frac{\partial (by \cdot bx)}{\partial y} + \frac{\partial (by \cdot 0)}{\partial z} \\ \frac{\partial (bx \cdot by)}{\partial x} + \frac{\partial (bx \cdot bx)}{\partial y} + \frac{\partial (bx \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot by)}{\partial x} + \frac{\partial (0 \cdot bx)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 + b^2 x + 0 \\ b^2 y + 0 + 0 \\ 0 + 0 + 0 \end{bmatrix} = \langle b^2 x, b^2 y, 0 \rangle$$

$$3c(2d) - \vec{v}_4 = \langle -by, bx, 0 \rangle$$

$$\mathbf{v}\mathbf{v} = \begin{bmatrix} -by & bx & 0 \end{bmatrix} \begin{bmatrix} -by \\ bx \\ 0 \end{bmatrix} = \begin{bmatrix} b^2 y^2 & -b^2 xy & 0 \\ -b^2 xy & b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot \mathbf{v}\mathbf{v} = \begin{bmatrix} \frac{\partial (-by \cdot -by)}{\partial x} + \frac{\partial (-by \cdot bx)}{\partial y} + \frac{\partial (-by \cdot 0)}{\partial z} \\ \frac{\partial (bx \cdot -by)}{\partial x} + \frac{\partial (bx \cdot bx)}{\partial y} + \frac{\partial (bx \cdot 0)}{\partial z} \\ \frac{\partial (0 \cdot -by)}{\partial x} + \frac{\partial (0 \cdot bx)}{\partial y} + \frac{\partial (0 \cdot 0)}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 - b^2 x + 0 \\ -b^2 y + 0 + 0 \\ 0 + 0 + 0 \end{bmatrix} = \langle -b^2 x, -b^2 y, 0 \rangle$$

9. If $\mathbf{r}$ is the position vector and r is its magnitude, verify that

(a) $\nabla \frac{1}{r} = -\frac{\mathbf{r}}{r^3}$ (c) $\nabla(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}$ if $\mathbf{a}$ is a constant vector

(b) $\nabla f(r) = \frac{1}{r} \frac{df}{dr} \mathbf{r}$

$$\mathbf{r} = \langle x, y, z \rangle \quad r = \sqrt{x^2 + y^2 + z^2}$$

$$9a - \nabla \frac{1}{r} = -\frac{\mathbf{r}}{r^3}$$

$$\begin{aligned} \nabla \left((x^2 + y^2 + z^2)^{-1/2} \right) &= \frac{\partial}{\partial x} \left((x^2 + y^2 + z^2)^{-1/2} \right), \frac{\partial}{\partial y} \left((x^2 + y^2 + z^2)^{-1/2} \right), \frac{\partial}{\partial z} \left((x^2 + y^2 + z^2)^{-1/2} \right) \\ &= 2x \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2}, \quad 2y \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2}, \quad 2z \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \\ &= -x (x^2 + y^2 + z^2)^{-3/2}, \quad -y (x^2 + y^2 + z^2)^{-3/2}, \quad -z (x^2 + y^2 + z^2)^{-3/2} \\ &= \left\langle \frac{-x}{(x^2 + y^2 + z^2)^{3/2}}, \frac{-y}{(x^2 + y^2 + z^2)^{3/2}}, \frac{-z}{(x^2 + y^2 + z^2)^{3/2}} \right\rangle = -\frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{r^3} \end{aligned}$$

$$-\frac{\mathbf{r}}{r^3} = -\frac{\langle x, y, z \rangle}{((x^2 + y^2 + z^2)^{3/2})} = -\frac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{3/2}}$$

$$9b - \nabla f(r) = \frac{1}{r} \frac{df}{dr} \mathbf{r} \quad \nabla f(r) = \frac{\partial}{\partial x} f(r) + \frac{\partial}{\partial y} f(r) + \frac{\partial}{\partial z} f(r)$$

$$\frac{\partial f}{\partial x} = \frac{df}{dr} \frac{\partial r}{\partial x} = \frac{df}{dr} \cdot 2x \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} = \frac{df}{dr} \cdot \frac{x}{r}$$

$$\frac{\partial f}{\partial y} = \frac{df}{dr} \frac{\partial r}{\partial y} = \frac{df}{dr} \cdot 2y \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} = \frac{df}{dr} \cdot \frac{y}{r}$$

$$\frac{\partial f}{\partial z} = \frac{df}{dr} \frac{\partial r}{\partial z} = \frac{df}{dr} \cdot 2z \cdot -\frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} = \frac{df}{dr} \cdot \frac{z}{r}$$

$$\begin{aligned} \nabla f(r) &= \left\langle \frac{df}{dr} \frac{x}{r}, \frac{df}{dr} \frac{y}{r}, \frac{df}{dr} \frac{z}{r} \right\rangle \\ &= \frac{df}{dr} \frac{1}{r} \langle x, y, z \rangle = \frac{df}{dr} \frac{1}{r} \mathbf{r} \end{aligned}$$

$$9c - \nabla(\mathbf{a} \cdot \mathbf{r}) = \mathbf{a}$$

$$\mathbf{a} = \langle a_x, a_y, a_z \rangle$$

$$\mathbf{r} = \langle x, y, z \rangle$$

$$\mathbf{a} \cdot \mathbf{r} = x a_x + y a_y + z a_z$$

$$\nabla(\mathbf{a} \cdot \mathbf{r}) = \frac{\partial}{\partial x} x a_x + \frac{\partial}{\partial y} y a_y + \frac{\partial}{\partial z} z a_z$$

$$= \langle a_x, a_y, a_z \rangle = \mathbf{a}$$

10. Write out in full in Cartesian coordinates

$$(a) \frac{\partial}{\partial t} \rho \mathbf{v} = -[\nabla \cdot \rho \mathbf{v} \mathbf{v}] - \nabla p - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g}$$

$$(b) \boldsymbol{\tau} = -\mu \{ \nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3} (\nabla \cdot \mathbf{v}) \boldsymbol{\delta} \}$$

$$\begin{aligned} \frac{\partial}{\partial t} \rho \mathbf{v} &= -[\nabla \cdot \rho \mathbf{v} \mathbf{v}] - \nabla p - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g} \\ &= -\left[\nabla \cdot \rho \begin{pmatrix} v_x & v_y & v_z \\ v_x & v_y & v_z \\ v_x & v_y & v_z \end{pmatrix} \right] - \left\langle \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z} \right\rangle - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g} \\ &= -\left[\nabla \cdot \rho \begin{pmatrix} v_{xx} & v_{xy} & v_{xz} \\ v_{yx} & v_{yy} & v_{yz} \\ v_{zx} & v_{zy} & v_{zz} \end{pmatrix} \right] - \left\langle \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z} \right\rangle - \begin{bmatrix} \frac{\partial \tau_{xx}}{\partial x} & \frac{\partial \tau_{xy}}{\partial y} & \frac{\partial \tau_{xz}}{\partial z} \\ \frac{\partial \tau_{yx}}{\partial x} & \frac{\partial \tau_{yy}}{\partial y} & \frac{\partial \tau_{yz}}{\partial z} \\ \frac{\partial \tau_{zx}}{\partial x} & \frac{\partial \tau_{zy}}{\partial y} & \frac{\partial \tau_{zz}}{\partial z} \end{bmatrix} + \rho \mathbf{g} \\ &= -\begin{bmatrix} \frac{\partial v_{xx}}{\partial x} & \frac{\partial v_{xy}}{\partial y} & \frac{\partial v_{xz}}{\partial z} \\ \frac{\partial v_{yx}}{\partial x} & \frac{\partial v_{yy}}{\partial y} & \frac{\partial v_{yz}}{\partial z} \\ \frac{\partial v_{zx}}{\partial x} & \frac{\partial v_{zy}}{\partial y} & \frac{\partial v_{zz}}{\partial z} \end{bmatrix} - \left\langle \frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z} \right\rangle - \begin{bmatrix} \frac{\partial \tau_{xx}}{\partial x} & \frac{\partial \tau_{xy}}{\partial y} & \frac{\partial \tau_{xz}}{\partial z} \\ \frac{\partial \tau_{yx}}{\partial x} & \frac{\partial \tau_{yy}}{\partial y} & \frac{\partial \tau_{yz}}{\partial z} \\ \frac{\partial \tau_{zx}}{\partial x} & \frac{\partial \tau_{zy}}{\partial y} & \frac{\partial \tau_{zz}}{\partial z} \end{bmatrix} + \rho \mathbf{g} \end{aligned}$$

$$\frac{\partial v_x}{\partial t} = -\left[\frac{\partial v_{xx}}{\partial x} + \frac{\partial v_{xy}}{\partial y} + \frac{\partial v_{xz}}{\partial z} \right] - \frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] + \rho g_x$$

$$\frac{\partial v_y}{\partial t} = -\left[\frac{\partial v_{yx}}{\partial x} + \frac{\partial v_{yy}}{\partial y} + \frac{\partial v_{yz}}{\partial z} \right] - \frac{\partial p}{\partial y} - \left[\frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right] + \rho g_y$$

$$\frac{\partial v_z}{\partial t} = -\left[\frac{\partial v_{zx}}{\partial x} + \frac{\partial v_{zy}}{\partial y} + \frac{\partial v_{zz}}{\partial z} \right] - \frac{\partial p}{\partial z} - \left[\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right] + \rho g_z$$

$$\boldsymbol{\tau} = -\mu \{ \nabla \mathbf{v} + (\nabla \mathbf{v})^T - \frac{2}{3} (\nabla \cdot \mathbf{v}) \boldsymbol{\delta} \}$$

$$\begin{bmatrix} \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = -\mu \left\{ \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix} + \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} - \frac{2}{3} (v_x + v_y + v_z) \right\}$$

$$\tau_{xy} = \tau_{yx} = -\mu (v_y + v_x)$$

$$\tau_{xz} = \tau_{zx} = -\mu (v_z + v_x)$$

$$\tau_{yz} = \tau_{zy} = -\mu (v_z + v_y)$$

§A.5 VECTOR AND TENSOR INTEGRAL THEOREMS

For performing general proofs in continuum physics, several integral theorems are extremely useful.

The Gauss–Ostrogradskii Divergence Theorem

If V is a closed region in space enclosed by a surface S , then

$$\int_V (\nabla \cdot \mathbf{v}) dV = \int_S (\mathbf{n} \cdot \mathbf{v}) dS \quad (\text{A.5-1})$$

in which $\mathbf{n}$ is the outwardly directed unit normal vector. This is known as the *divergence theorem* of Gauss and Ostrogradskii. Two closely allied theorems for scalars and tensors are

1. Consider the vector field

$$\mathbf{v} = \delta_1 x_1 + \delta_2 x_3 + \delta_3 x_2$$

Evaluate both sides of Eq. A.5-1 over the region bounded by the planes $x_1 = 0$, $x_1 = 1$; $x_2 = 0$, $x_2 = 2$; $x_3 = 0$, $x_3 = 4$.

$$\int_V (\nabla \cdot \mathbf{v}) dV = \int_S (\mathbf{n} \cdot \mathbf{v}) dS$$

$\mathbf{n}$ is the unit normal vector

$$\mathbf{v} = \delta_1 x_1 + \delta_2 x_3 + \delta_3 x_2$$

$$= \langle x_1, x_3, x_2 \rangle$$

$$\nabla \cdot \mathbf{v} = \frac{\partial x_1}{\partial x_1} + \frac{\partial x_3}{\partial x_2} + \frac{\partial x_2}{\partial x_3}$$

$$= 1$$

$$\int_0^1 \int_0^2 \int_0^4 1 \, dx_1 dx_2 dx_3 = \int_0^4 \int_0^2 \int_0^1 1 \, dx_1 dx_2 dx_3 = 4 \cdot 2 \cdot 1 = 8$$

$$x_1 = \langle 1, 0, 0 \rangle$$

$$\mathbf{n} = \langle 1, 0, 0 \rangle$$

$$x_1 \cdot \mathbf{n} = 1 \cdot 1 + 0 + 0 = 1 = x_1$$

$$\int_0^2 \int_0^4 1 \, dx_2 dx_3 = \int_0^4 \int_0^2 1 \, dx_2 dx_3 = 4 \cdot 2 = 8$$

Cylindrical Coordinates

From Eqs. A.6-10, 11, and 12 we can obtain expressions for the spatial derivatives of the unit vectors δ_r , δ_θ , and δ_z :

$$\frac{\partial}{\partial r} \delta_r = 0 \quad \frac{\partial}{\partial r} \delta_\theta = 0 \quad \frac{\partial}{\partial r} \delta_z = 0 \quad (\text{A.7-1})$$

$$\frac{\partial}{\partial \theta} \delta_r = -\delta_\theta \quad \frac{\partial}{\partial \theta} \delta_\theta = -\delta_r \quad \frac{\partial}{\partial \theta} \delta_z = 0 \quad (\text{A.7-2})$$

$$\frac{\partial}{\partial z} \delta_r = 0 \quad \frac{\partial}{\partial z} \delta_\theta = 0 \quad \frac{\partial}{\partial z} \delta_z = 0 \quad (\text{A.7-3})$$

1. If $\mathbf{r}$ is the instantaneous position vector for a particle, show that the velocity and acceleration of the particle are given by (use Eq. A.7-2):

$$\mathbf{v} = \frac{d}{dt} \mathbf{r} = \delta_r \dot{r} + \delta_\theta r \dot{\theta} + \delta_z \dot{z} \quad (\text{A.7-34})$$

$$\mathbf{a} = \delta_r (\ddot{r} - r\dot{\theta}^2) + \delta_\theta (r\ddot{\theta} + 2\dot{r}\dot{\theta}) + \delta_z \ddot{z} \quad (\text{A.7-35})$$

in cylindrical coordinates. The dots indicate time derivatives of the coordinates.

$$\mathbf{r} = \frac{d\mathbf{r}}{dt} = \delta_r \frac{dr}{dt} + \delta_\theta \frac{d\theta}{dt} + \delta_z \frac{dz}{dt}$$

$$\begin{aligned} \mathbf{r} &= r\delta_r + z\delta_z \\ \mathbf{v} &= \frac{d}{dt} \mathbf{r} = \frac{d}{dt} (r\delta_r) + \frac{d}{dt} (z\delta_z) \\ &= \delta_r \frac{dr}{dt} + r \frac{d\delta_r}{dt} + \delta_z \frac{dz}{dt} + z \frac{d\delta_z}{dt} \\ &= \dot{r}\delta_r + r\dot{\delta}_r + \dot{z}\delta_z + z\dot{\delta}_z \\ &= \dot{r}\delta_r + r\dot{\theta}\delta_\theta + \dot{z}\delta_z \quad \frac{d\delta_r}{dt} = \frac{d\delta_r}{d\theta} \frac{d\theta}{dt} \end{aligned}$$

$$\frac{d\delta_r}{dt} = \frac{\partial \delta_r}{\partial \theta} \frac{\partial \theta}{\partial t} = \theta \frac{d\delta_\theta}{dt} = \theta \delta_\theta$$

$$\begin{aligned} \mathbf{a} &= \frac{d}{dt} \mathbf{v} = \frac{d^2 \mathbf{r}}{dt^2} = \frac{d}{dt} (\dot{r}\delta_r) + \frac{d}{dt} (r\dot{\theta}\delta_\theta) + \frac{d}{dt} (\dot{z}\delta_z) \\ &= \ddot{r}\delta_r + \dot{r}\dot{\delta}_r + (\dot{r}\dot{\theta} + r\ddot{\theta})\delta_\theta + (r\ddot{\theta})\delta_\theta + \dot{z}\delta_z \\ &= \ddot{r}\delta_r + r(\dot{\theta}\delta_\theta) + (\dot{r}\dot{\theta} + r\ddot{\theta})\delta_\theta + [(r\ddot{\theta})(-\dot{\theta}\delta_r)] + \dot{z}\delta_z \\ &= \ddot{r}\delta_r + r\dot{\theta}\delta_\theta + \dot{r}\dot{\theta}\delta_\theta + r\ddot{\theta}\delta_\theta - r\dot{\theta}^2\delta_r + \dot{z}\delta_z \\ &= \ddot{r}\delta_r - r\dot{\theta}^2\delta_r + (2\dot{r}\dot{\theta} + r\ddot{\theta})\delta_\theta + \dot{z}\delta_z \\ &= \delta_r (\ddot{r} - r\dot{\theta}^2) + \delta_\theta (2\dot{r}\dot{\theta} + r\ddot{\theta}) + \delta_z \ddot{z} \end{aligned}$$

3. Use Table A.7-2 to write down directly the following quantities in cylindrical coordinates:

- (a) $(\nabla \cdot \rho \mathbf{v})$, where ρ is a scalar (b) $[\nabla \cdot \rho \mathbf{v} \mathbf{v}]_r$, where ρ is a scalar
(c) $[\nabla \cdot p \delta]_\theta$, where p is a scalar (d) $(\nabla \cdot [\boldsymbol{\tau} \cdot \mathbf{v}])$
(e) $[\mathbf{v} \cdot \nabla \mathbf{v}]_\theta$ (f) $\nabla \mathbf{v} + (\nabla \mathbf{v})^\dagger$

$$3a - (\nabla \cdot \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \quad \mathbf{v} = v_r \delta_r + v_\theta \delta_\theta + v_z \delta_z$$

$$(\nabla \cdot \rho \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (\rho r v_r) + \frac{1}{r} \frac{\partial (\rho v_\theta)}{\partial \theta} + \frac{\partial (\rho v_z)}{\partial z}$$

$$3b - [\nabla \cdot \rho \mathbf{v} \mathbf{v}]_r = \rho \mathbf{v} \mathbf{v} = \rho \begin{bmatrix} v_r & v_\theta & v_z \\ v_\theta & v_r & v_z \\ v_z & v_r & v_\theta \end{bmatrix} = \begin{bmatrix} \rho v_r^2 & \rho v_r v_\theta & \rho v_r v_z \\ \rho v_\theta v_r & \rho v_\theta^2 & \rho v_\theta v_z \\ \rho v_z v_r & \rho v_z v_\theta & \rho v_z^2 \end{bmatrix}$$

$$\text{Using 1} \Rightarrow [\nabla \cdot \boldsymbol{\tau}]_r = \frac{1}{r} \frac{\partial}{\partial r} (r \rho v_r^2) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta v_r) + \frac{\partial}{\partial z} (\rho v_z v_r) - \frac{\rho v_\theta^2}{r}$$

$$3c - [\nabla \cdot p \delta]_\theta = \frac{1}{r} \frac{\partial}{\partial \theta} p$$

$$p \delta = \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix} \quad \text{Using 1} \Rightarrow [\nabla \cdot \boldsymbol{\tau}]_\theta = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r} \frac{\partial}{\partial \theta} \tau_{\theta\theta} + \frac{\partial}{\partial z} \tau_{z\theta} + \frac{\tau_{\theta r} - \tau_{r\theta}}{r}$$

$$= \frac{1}{r} \frac{\partial \tau_{\theta\theta}}{\partial \theta} = \frac{1}{r} \frac{\partial p}{\partial \theta}$$

$$3d - (\nabla \cdot [\boldsymbol{\tau} \cdot \mathbf{v}])$$

$$[\boldsymbol{\tau} \cdot \mathbf{v}] = \begin{pmatrix} \tau_{rr} v_r + \tau_{r\theta} v_\theta + \tau_{rz} v_z, \tau_{\theta r} v_r + \tau_{\theta\theta} v_\theta + \tau_{\theta z} v_z, \tau_{zr} v_r + \tau_{z\theta} v_\theta + \tau_{zz} v_z \end{pmatrix}$$

$$\begin{bmatrix} \tau_{rr} & \tau_{r\theta} & \tau_{rz} \\ \tau_{\theta r} & \tau_{\theta\theta} & \tau_{\theta z} \\ \tau_{zr} & \tau_{z\theta} & \tau_{zz} \end{bmatrix} \begin{bmatrix} v_r \\ v_\theta \\ v_z \end{bmatrix}$$

$$\text{Using A} - (\nabla \cdot \mathbf{v}) = \frac{1}{r} \frac{\partial}{\partial r} (r (\tau_{rr} v_r + \tau_{r\theta} v_\theta + \tau_{rz} v_z)) + \frac{1}{r} \frac{\partial}{\partial \theta} (\tau_{\theta r} v_r + \tau_{\theta\theta} v_\theta + \tau_{\theta z} v_z) + \frac{\partial}{\partial z} (\tau_{zr} v_r + \tau_{z\theta} v_\theta + \tau_{zz} v_z)$$

$$3e - [\mathbf{v} \cdot \nabla \mathbf{v}]_\theta = \begin{bmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{\partial v_r}{\partial \theta} & \frac{\partial v_\theta}{\partial \theta} & \frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix} \begin{bmatrix} v_r & v_\theta & v_z \end{bmatrix} = \begin{bmatrix} \frac{\partial v_r}{\partial r} v_r & \frac{\partial v_r}{\partial \theta} v_\theta & \frac{\partial v_r}{\partial z} v_z \\ \frac{\partial v_\theta}{\partial r} v_r & \frac{\partial v_\theta}{\partial \theta} v_\theta & \frac{\partial v_\theta}{\partial z} v_z \\ \frac{\partial v_z}{\partial r} v_r & \frac{\partial v_z}{\partial \theta} v_\theta & \frac{\partial v_z}{\partial z} v_z \end{bmatrix} = v_r \frac{\partial v_\theta}{\partial r} + v_\theta \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + v_z \frac{\partial v_\theta}{\partial z}$$

$$3f - \nabla \mathbf{v} + (\nabla \mathbf{v})^\dagger = \begin{bmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{1}{r} \frac{\partial v_r}{\partial \theta} & \frac{v_r}{r} & \frac{1}{r} \frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

$$\nabla \mathbf{v} = \begin{bmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial r} & \frac{\partial v_z}{\partial r} \\ \frac{\partial v_r}{\partial \theta} & \frac{\partial v_\theta}{\partial \theta} & \frac{\partial v_z}{\partial \theta} \\ \frac{\partial v_r}{\partial z} & \frac{\partial v_\theta}{\partial z} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

$$\nabla \mathbf{v}^\dagger = \begin{bmatrix} \frac{\partial v_r}{\partial r} & \frac{\partial v_\theta}{\partial \theta} & \frac{\partial v_z}{\partial z} \\ \frac{\partial v_\theta}{\partial r} & \frac{\partial v_\theta}{\partial \theta} & \frac{\partial v_\theta}{\partial z} \\ \frac{\partial v_z}{\partial r} & \frac{\partial v_z}{\partial \theta} & \frac{\partial v_z}{\partial z} \end{bmatrix}$$

$$u - \text{div}(\rho \langle v \rangle) = \langle \nabla \rho | v \rangle + \rho \text{div} \langle v \rangle$$

$$\nabla \rho \cdot v \quad \rho \langle \nabla \cdot v \rangle \rightarrow \frac{\partial \rho}{\partial x} \cdot v_x + \rho \frac{\partial v_x}{\partial x} + \rho \frac{\partial v_y}{\partial x}$$

$$\rho(v) = \langle \rho v_x, \rho v_y, \rho v_z \rangle$$

$$\left(\frac{\partial \rho}{\partial x}, \frac{\partial \rho}{\partial y}, \frac{\partial \rho}{\partial z} \right) \cdot \langle v_x, v_y, v_z \rangle$$

$$v_x \frac{\partial \rho}{\partial x} + v_y \frac{\partial \rho}{\partial y} + v_z \frac{\partial \rho}{\partial z}$$

$$\text{div}(\rho \langle v \rangle) = \frac{\partial \rho v_x}{\partial x} + \frac{\partial \rho v_y}{\partial y} + \frac{\partial \rho v_z}{\partial z}$$

$$= \left(v_x \frac{\partial \rho}{\partial x} + \rho \frac{\partial v_x}{\partial x} \right) + \left(v_y \frac{\partial \rho}{\partial y} + \rho \frac{\partial v_y}{\partial y} \right) + \left(v_z \frac{\partial \rho}{\partial z} + \rho \frac{\partial v_z}{\partial z} \right)$$

$$= \left(v_x \frac{\partial \rho}{\partial x} + v_y \frac{\partial \rho}{\partial y} + v_z \frac{\partial \rho}{\partial z} \right) + \left[\rho \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \right]$$

$$= (\nabla \rho \cdot v) + \rho (\nabla \cdot v)$$

$$b - \text{div}(w) \langle w \rangle = |w| w + \langle w w \rangle |w|$$

$$\nabla(w w) = (\nabla w \cdot w) + (\nabla \cdot w) w$$

$$v = \langle v_x, v_y, v_z \rangle$$

$$w = \langle w_x, w_y, w_z \rangle$$

$$\begin{bmatrix} v_x & v_y & v_z \\ w_x & w_y & w_z \end{bmatrix} = \begin{bmatrix} v_x w_x & v_x w_y & v_x w_z \\ v_y w_x & v_y w_y & v_y w_z \\ v_z w_x & v_z w_y & v_z w_z \end{bmatrix}$$

$$\nabla \begin{bmatrix} v_x w_x & v_x w_y & v_x w_z \\ v_y w_x & v_y w_y & v_y w_z \\ v_z w_x & v_z w_y & v_z w_z \end{bmatrix} = \left\langle \begin{array}{l} \frac{\partial v_x w_x}{\partial x} + \frac{\partial v_x w_y}{\partial y} + \frac{\partial v_x w_z}{\partial z}, \\ \frac{\partial v_y w_x}{\partial x} + \frac{\partial v_y w_y}{\partial y} + \frac{\partial v_y w_z}{\partial z}, \\ \frac{\partial v_z w_x}{\partial x} + \frac{\partial v_z w_y}{\partial y} + \frac{\partial v_z w_z}{\partial z} \end{array} \right\rangle$$

$$\left[\frac{\partial v_x}{\partial x}, \frac{\partial v_y}{\partial y}, \frac{\partial v_z}{\partial z} \right] \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} = \begin{bmatrix} \frac{\partial v_x}{\partial x} w_x + \frac{\partial v_y}{\partial y} w_x + \frac{\partial v_z}{\partial z} w_x \\ \frac{\partial v_x}{\partial x} w_y + \frac{\partial v_y}{\partial y} w_y + \frac{\partial v_z}{\partial z} w_y \\ \frac{\partial v_x}{\partial x} w_z + \frac{\partial v_y}{\partial y} w_z + \frac{\partial v_z}{\partial z} w_z \end{bmatrix}$$

$$\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) \cdot \langle v_x, v_y, v_z \rangle = v_x \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) + v_y \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right) + v_z \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right)$$

$$\left\langle \frac{\partial v_x w_x}{\partial x} + \frac{\partial v_x w_y}{\partial y} + \frac{\partial v_x w_z}{\partial z}, \frac{\partial v_y w_x}{\partial x} + \frac{\partial v_y w_y}{\partial y} + \frac{\partial v_y w_z}{\partial z}, \frac{\partial v_z w_x}{\partial x} + \frac{\partial v_z w_y}{\partial y} + \frac{\partial v_z w_z}{\partial z} \right\rangle$$

$$w_x \frac{\partial v_x}{\partial x} + v_x \frac{\partial w_x}{\partial x} + w_y \frac{\partial v_x}{\partial y} + v_y \frac{\partial w_x}{\partial y} + w_z \frac{\partial v_x}{\partial z} + v_z \frac{\partial w_x}{\partial z},$$

$$w_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial w_x}{\partial x} + w_y \frac{\partial v_y}{\partial y} + v_y \frac{\partial w_y}{\partial y} + w_z \frac{\partial v_y}{\partial z} + v_z \frac{\partial w_y}{\partial z}$$

$$w_x \frac{\partial v_z}{\partial x} + v_z \frac{\partial w_x}{\partial x} + w_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial w_y}{\partial y} + w_z \frac{\partial v_z}{\partial z} + v_z \frac{\partial w_z}{\partial z}$$

$$\left(w_x \frac{\partial v_x}{\partial x} + w_y \frac{\partial v_x}{\partial y} + w_z \frac{\partial v_x}{\partial z} + v_x \left[\frac{\partial w_x}{\partial x} + \frac{\partial w_y}{\partial y} + \frac{\partial w_z}{\partial z} \right] + w_x \frac{\partial v_y}{\partial x} + w_y \frac{\partial v_y}{\partial y} + w_z \frac{\partial v_y}{\partial z} + v_y \left[\frac{\partial w_x}{\partial x} + \frac{\partial w_y}{\partial y} + \frac{\partial w_z}{\partial z} \right] + w_x \frac{\partial v_z}{\partial x} + w_y \frac{\partial v_z}{\partial y} + w_z \frac{\partial v_z}{\partial z} + v_z \left[\frac{\partial w_x}{\partial x} + \frac{\partial w_y}{\partial y} + \frac{\partial w_z}{\partial z} \right] \right) = (\nabla v) w + (v \cdot \nabla) w$$

$$c - \text{div}(v | \tau |) = | \tau | | \nabla v | + \langle v | \text{div} \tau \rangle$$

$$\nabla \cdot (v \cdot \tau) = (\tau : \nabla v) + v \cdot (\nabla \cdot \tau)$$

$$\begin{bmatrix} v_x & v_y & v_z \\ \tau_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \tau_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \tau_{zz} \end{bmatrix} = \begin{bmatrix} v_x \tau_{xx} & v_y \tau_{xy} & v_z \tau_{xz} \\ v_x \tau_{yx} & v_y \tau_{yy} & v_z \tau_{yz} \\ v_x \tau_{zx} & v_y \tau_{zy} & v_z \tau_{zz} \end{bmatrix}$$

$$\nabla \begin{bmatrix} v_x \tau_{xx} & v_y \tau_{xy} & v_z \tau_{xz} \\ v_x \tau_{yx} & v_y \tau_{yy} & v_z \tau_{yz} \\ v_x \tau_{zx} & v_y \tau_{zy} & v_z \tau_{zz} \end{bmatrix} = \begin{bmatrix} \frac{\partial v_x \tau_{xx}}{\partial x} + \frac{\partial v_y \tau_{xy}}{\partial y} + \frac{\partial v_z \tau_{xz}}{\partial z}, \\ \frac{\partial v_x \tau_{yx}}{\partial x} + \frac{\partial v_y \tau_{yy}}{\partial y} + \frac{\partial v_z \tau_{yz}}{\partial z}, \\ \frac{\partial v_x \tau_{zx}}{\partial x} + \frac{\partial v_y \tau_{zy}}{\partial y} + \frac{\partial v_z \tau_{zz}}{\partial z} \end{bmatrix} = \begin{bmatrix} \tau_{xx} \frac{\partial v_x}{\partial x} + v_x \frac{\partial \tau_{xx}}{\partial x} + \tau_{xy} \frac{\partial v_y}{\partial y} + v_y \frac{\partial \tau_{xy}}{\partial y} + \tau_{xz} \frac{\partial v_z}{\partial z} + v_z \frac{\partial \tau_{xz}}{\partial z} \\ \tau_{yx} \frac{\partial v_x}{\partial x} + v_x \frac{\partial \tau_{yx}}{\partial x} + \tau_{yy} \frac{\partial v_y}{\partial y} + v_y \frac{\partial \tau_{yy}}{\partial y} + \tau_{yz} \frac{\partial v_z}{\partial z} + v_z \frac{\partial \tau_{yz}}{\partial z} \\ \tau_{zx} \frac{\partial v_x}{\partial x} + v_x \frac{\partial \tau_{zx}}{\partial x} + \tau_{zy} \frac{\partial v_y}{\partial y} + v_y \frac{\partial \tau_{zy}}{\partial y} + \tau_{zz} \frac{\partial v_z}{\partial z} + v_z \frac{\partial \tau_{zz}}{\partial z} \end{bmatrix}$$

$$= \left[\tau_{xx} \frac{\partial v_x}{\partial x} + \tau_{xy} \frac{\partial v_y}{\partial y} + \tau_{xz} \frac{\partial v_z}{\partial z}, \tau_{yx} \frac{\partial v_x}{\partial x} + \tau_{yy} \frac{\partial v_y}{\partial y} + \tau_{yz} \frac{\partial v_z}{\partial z}, \tau_{zx} \frac{\partial v_x}{\partial x} + \tau_{zy} \frac{\partial v_y}{\partial y} + \tau_{zz} \frac{\partial v_z}{\partial z} \right] + \left[v_x \frac{\partial \tau_{xx}}{\partial x} + v_y \frac{\partial \tau_{xy}}{\partial y} + v_z \frac{\partial \tau_{xz}}{\partial z}, v_x \frac{\partial \tau_{yx}}{\partial x} + v_y \frac{\partial \tau_{yy}}{\partial y} + v_z \frac{\partial \tau_{yz}}{\partial z}, v_x \frac{\partial \tau_{zx}}{\partial x} + v_y \frac{\partial \tau_{zy}}{\partial y} + v_z \frac{\partial \tau_{zz}}{\partial z} \right]$$

$$= (\tau : \nabla v) + v \cdot (\nabla \cdot \tau)$$

$$= | \tau | | \nabla v | + \langle v | \text{div} \tau \rangle$$